

Challenge (Habanero)

- Q1.** A hiker is traveling at a ratio of 3 : 5 miles to hours. If the hiker travels for 2 hours, how many miles will they have traveled?
(A) 1.2 miles
(B) 3.6 miles
(C) 2.4 miles
(D) 1.8 miles
(E) 2.2 miles
- Q2.** A ratio of $x : y$ represents the number of birds to the number of trees in a forest. If $x : y = 4 : 7$, and there are 28 trees, how many birds are in the forest?
(A) 12 birds
(B) 16 birds
(C) 10 birds
(D) 8 birds
(E) 14 birds
- Q3.** A car's fuel consumption is 25 : 1 (miles to gallons). If the car travels 250 miles, how much fuel will it have consumed?
(A) 8 gallons
(B) 10 gallons
(C) 12 gallons
(D) 6 gallons
(E) 5 gallons
- Q4.** A ratio of $a : b$ represents the number of employees to the number of computers in an office. If $a : b = 3 : 5$ and there are 15 employees, how many computers are in the office?
(A) 25 computers
(B) 20 computers
(C) 15 computers
(D) 10 computers
- (E) 8 computers
- Q5.** A group of friends want to share some food in the ratio 2 : 3 : 5. If they have 50 pieces of food, how many pieces will the person with ratio 5 get?
(A) 20 pieces
(B) 25 pieces
(C) 15 pieces
(D) 18 pieces
(E) 12 pieces
- Q6.** A park has a ratio of 2 : 3 (park rangers to visitors). If there are 36 visitors, how many park rangers are there?
(A) 18 rangers
(B) 24 rangers
(C) 12 rangers
(D) 16 rangers
(E) 8 rangers
- Q7.** A bicycle's gear ratio is 4 : 7. If the front gear has 28 teeth, how many teeth does the back gear have?
(A) 40 teeth
(B) 49 teeth
(C) 32 teeth
(D) 35 teeth
(E) 24 teeth
- Q8.** A farmer has a ratio of 3 : 2 (cows to pigs). If there are 12 pigs, how many cows does the farmer have?
(A) 12 cows
(B) 16 cows
(C) 18 cows
(D) 8 cows
(E) 9 cows

Open Ended Questions (FRQ)

- Q9.** A group of hikers are traveling through the wilderness. Their map has a scale of 1 : 25000. If two villages are 5 cm apart on the map, how far apart are they in reality? Show your work and express your answer in kilometers.
- Q10.** A zoo has a ratio of 2 : 5 (lions to tigers). If there are 30 tigers, and the zoo wants to maintain the ratio, how many lions should they have? If they currently have 10 lions, how many more lions should they get to maintain the ratio? Show your work and express your answers.

Chef's Tips

- Remind students to simplify ratios when possible
- Encourage students to use cross-multiplication when solving proportions
- Have students check their units when solving problems involving real-world applications